CO2-to-Fuel Conversion Feasibility Study

Case Competition Submission — Sample Report

Prepared by Jordan Reyes | Sample project summary for portfolio viewing

OVERVIEW

This sample report summarizes a proposal evaluating the technical and economic feasibility of converting captured CO_2 and renewable hydrogen into synthetic fuels using existing natural gas infrastructure for storage and transport.

APPROACH

A process flow model was built to estimate reactor sizing, catalyst requirements, and utility loads. A techno-economic assessment compared production costs against conventional fuel pathways under several hydrogen price scenarios.

RESULTS

The proposal placed first in a 24-team national case competition, with judges highlighting the clarity of the economic sensitivity analysis.

NOTES

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